

CommunityTwin: An AI-Driven Digital Twin Framework for Water Shortage and Flood Risk Prediction in Communities

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ABSTRACT Rural and urban communities struggle with water shortages, flooding and waste accumulation, yet these problems are usually monitored in separate systems, which prevents a unified view and makes it hard to decide which issue to address first. This paper presents CommunityTwin, a software-based, AI-powered digital twin that fuses simulated sensor data, citizen reports, weather, geographical and historical data into a single community model. Machine-learning modules predict village-level water deficit and ward-level flood risk, and a priority engine ranks locations by combined risk and exposure. A what-if simulation module estimates the effect of interventions such as improved drainage or heavier rainfall. On simulated data, logistic regression reached an ROC-AUC of 0.961 and F1 of 0.766 for flood risk, tree ensembles reached 0.943-0.949, and linear regression reached R^2 of 0.962 for water deficit prediction. Increasing drainage capacity by 25% and removing waste blockage reduced mean predicted flood probability from 0.259 to 0.155. Results are based on synthetic data and demonstrate pipeline feasibility rather than field performance.

INDEX TERMS Digital twin, community resilience, water shortage prediction, flood risk prediction, priority engine, what-if simulation, machine learning, smart community.

I. INTRODUCTION

Communities of every size, from small villages to large urban wards, depend on three basic services: a reliable supply of water, protection from flooding and waterlogging, and the regular collection of solid waste. When any of these fails, the consequences spread. Households without sufficient water store it unsafely or travel long distances to collect it. Streets that flood expose residents to contaminated water, interrupt transport and damage property. Uncollected waste attracts disease vectors and, importantly, blocks the drains and channels that are meant to carry storm water away. In practice these problems are interlinked: blocked drains aggravate flooding, flooding contaminates water sources, and heat and population growth increase demand on supplies that are already stressed.

Despite this interdependence, the information systems that support local administration are usually organised by service. A water department may keep records of tank levels and tanker deliveries, a drainage or irrigation office may track rainfall and river gauges, and a sanitation department may maintain a complaint register for waste collection. Citizen complaints arrive through telephone calls, paper registers, field visits and, increasingly, mobile applications, and are generally handled independently of one another. Decision makers therefore lack a unified view of the community. They cannot easily see where problems cluster, how one problem affects another, or, most importantly, which problem deserves attention first when staff, vehicles and budget are limited.

A digital twin (DT) is a virtual representation of a physical system that is kept up to date with data from that system and is used for analysis and decision support [1], [2]. The concept has matured in manufacturing, where a twin of a machine or production line is synchronised with sensor streams and used for monitoring and optimisation [3]. More recently it has been applied to smart cities, where twins integrate building

information models (BIM) and geographic information system (GIS) data [5]. Most existing work, however, targets a single asset or a single service. Few studies describe a lightweight, software-only twin of an entire community that combines hazard prediction with prioritisation and scenario testing in a form that a non-technical administrator can use.

Recent AI and Internet-of-Things (IoT) frameworks have improved water management through ensemble learning and edge deployment [4]. Such frameworks, however, focus on one resource and assume that dense physical sensor networks already exist. Many rural communities, and many urban neighbourhoods in low- and middle-income settings, have no such infrastructure. A practical solution must therefore be able to start from whatever data are available (a few gauges, weather forecasts, elevation and drainage maps, historical records and citizen reports) and to accept better data, such as real sensor streams, as they become available without redesign.

This paper presents **CommunityTwin**, a software-based, AI-powered digital twin that works from simulated sensor data, citizen reports, citizen-submitted images, weather and geographical data, and historical records. CommunityTwin represents each ward or village as a compact state vector, predicts village-level water deficit and ward-level flood risk with interpretable machine-learning models, ranks wards by combined risk and exposure in a priority engine, and estimates the effect of interventions in a what-if module. Because no field deployment has yet been carried out, all quantitative evidence in this paper comes from simulated data. The purpose of the evaluation is therefore to demonstrate that the pipeline is complete, internally consistent and reproducible, and to expose its sensitivities, rather than to claim predictive accuracy for any real community.

The main contributions of this paper are as follows:

- 1 A layered architecture that follows the principle *Monitor, Understand, Predict, Prioritise, Simulate, Act* and unifies heterogeneous community data in a single twin.
- 2 Machine-learning models for village-level water shortage prediction and ward-level flood risk prediction, evaluated on simulated data with reproducible protocols, including cross-validation and a feature-ablation study.
- 3 A priority engine and a what-if simulation facility that turn model outputs into ranked, actionable recommendations, together with a sensitivity analysis of the priority weights.
- 4 An honest evaluation, including the observation that simple linear models can match or outperform tree ensembles when the underlying relationships are close to linear, and a discussion of the limitations of simulated data.

The remainder of the paper is organised as follows. Section II reviews related work. Section III formalises the problem and describes the architecture. Section IV details the methodology, and Section V the experimental set-up. Section VI reports and discusses results. Section VII discusses implications for deployment, Section VIII considers ethical and societal issues, Section IX lists limitations and threats to validity, Section X outlines future research directions, and Section XI concludes.

II. RELATED WORK

A. Digital Twin Technology

The term *digital twin* was popularised in the context of product lifecycle management, where Grieves and Vickers [1] described a virtual counterpart of a physical product that mirrors its state throughout its life and can be used to anticipate unwanted emergent behaviour. Tao et al. [3] review the state of the art in industrial DTs and stress the fusion of physical and virtual data as the defining capability. Fuller et al. [2] survey the enabling technologies, challenges and open research questions of DTs across domains and highlight recurring obstacles: data quality, interoperability between heterogeneous sources, security and privacy, and the lack of standard reference architectures. These obstacles apply with particular force to community-scale twins, where data come from many organisations in inconsistent formats.

B. Digital Twins for Cities and Communities

Smart-city twins are typically constructed from BIM and GIS models that describe buildings, roads and utilities in great geometric detail. Jin et al. [5] show that such models become heavy and slow when served to web applications, and propose a lightweighting process so that numerous real-time services can run on the web. CommunityTwin adopts the same lightweight philosophy but takes it further. Instead of reducing the geometry of a detailed model, it never builds one: each ward or village is represented by a small vector of state variables (rainfall, drainage condition, elevation, water availability, waste reports and population), which is sufficient for the decision problems addressed here and cheap to store, update and serve.

C. AI for Water Management

Rokaya Mahmoud et al. [4] combine IoT data with ensemble models (XGBoost and LightGBM), metaheuristic feature selection and model quantisation for irrigation scheduling and leak detection, and show that AI-based forecasting is viable at the edge. Their work illustrates both the promise of AI for water and the typical scope of current studies: a single water use case, supported by a well-instrumented physical network. We extend this line of work from a single use case to a multi-hazard community view, and we deliberately rely on lighter, interpretable models whose behaviour an administrator can inspect.

D. Hazard Prediction and Spatial Analysis

Random forests [6], gradient boosting [7] and recurrent networks [8] are widely used for hydrological prediction, because they can capture non-linear interactions between rainfall, antecedent moisture, topography and channel conditions. Recurrent architectures such as long short-term memory networks [8] are particularly attractive when long time series are available. In the present work we restrict ourselves to tabular models trained on daily or weekly records, which is a more realistic starting point for communities with short or patchy histories; LSTM-based forecasting is listed as future work. Density-based clustering such as DBSCAN [10] is well suited to detecting spatial hotspots in geotagged events, because it finds arbitrarily shaped dense regions and labels isolated points as noise. We use it for waste analysis, which is reported in detail in a companion paper focused on waste hotspot detection, priority ranking and what-if simulation.

Beyond purely data-driven methods, process-based hydrological and hydraulic models simulate the movement of water through catchments and drainage networks from physical equations. They can represent the effect of a new culvert or a widened channel explicitly, which statistical models cannot, but they require detailed survey data, calibration against observed flows and specialist expertise. Data-driven models trade this physical fidelity for speed, simplicity and the ability to learn from whatever records exist. A mature community twin is likely to combine the two, using fast statistical models for continuous monitoring and screening, and process-based models to verify the few interventions that merit capital investment. CommunityTwin positions itself at the first of these levels.

Citizen participation is a further strand of related work that the twin builds on. Geotagged complaints, photographs and mobile reporting applications turn residents into distributed sensors, and are especially valuable where physical sensing is sparse. Their well-known weaknesses are uneven coverage, duplicate and malicious reports, and variable quality, which is why CommunityTwin applies duplicate removal and, for the waste module, image screening and density-based clustering before reports are allowed to influence a ranking.

E. Multi-Criteria Prioritisation

Ranking locations for intervention is a multi-criteria decision problem. Weighted-sum scoring and the analytic hierarchy process are widely used for such problems because they are transparent and can be negotiated with stakeholders. We adopt a simple weighted sum with an explicit exposure multiplier, whose weights are stored as configuration and can be altered by administrators, and we analyse how sensitive the resulting ranking is to those weights.

F. Positioning

Table 1 summarises how CommunityTwin differs from the studies discussed above. In short, the surveyed works either review the DT concept [2], [3], address a single service or asset [4], [5], or depend on dense sensing or detailed 3-D models. CommunityTwin addresses multiple hazards together, works from sparse and simulated data, and includes decision-support functions (prioritisation and what-if analysis) that the other works do not treat as a focus.

TABLE 1. COMPARISON OF COMMUNITYTWIN WITH THE RELATED STUDIES DISCUSSED IN THIS PAPER.

Study	Main focus	Data basis	Multi-hazard	Priority / what-if
Fuller et al. [2]	DT survey	Literature	N/A	N/A
Tao et al. [3]	DT in industry	Literature	N/A	N/A
Jin et al. [5]	Lightweight smart-city DT	BIM-GIS	No	Not a focus
Rokaya Mahmoud et al. [4]	Irrigation, leak detection	IoT sensors	No (water only)	Not a focus
CommunityTwin in (this work)	Community DT: water, flood, waste	Sim. sensors, reports, images, weather, GIS	Yes	Yes

III. PROBLEM FORMULATION AND ARCHITECTURE

A. Problem Formulation

Let a community be partitioned into a set of spatial units U (wards for flood and waste analysis, villages for water analysis). Time is discretised into days for flood analysis and weeks for water analysis. For each unit i and time step t the twin maintains a state vector $\mathbf{x}_{i,t}$ assembled from the data sources described below. Four decision problems are defined on this state:

- **Water shortage prediction.** Estimate the deficit ratio $D_{i,t} = (\text{demand} - \text{supply}) / \text{demand}$ for each village-week and flag the village as at risk when D exceeds the 75th percentile of the observed distribution.
- **Flood risk prediction.** Estimate the probability that ward i experiences a flood event on day t , and issue an alert when the probability exceeds a configurable threshold.
- **Prioritisation.** Combine the normalised risk scores of the three hazards with an exposure factor to produce a single ranking of wards for intervention.
- **Scenario analysis.** Given a hypothetical change to the state (for example more drainage capacity), estimate the resulting change in predicted risk.

The first two problems are supervised learning tasks; the third is a deterministic scoring rule; the fourth re-uses the trained models on a modified state. Keeping the pipeline in this form has a practical advantage: every step can be inspected, re-run and audited by a non-specialist.

B. Architecture Overview

Fig. 1 shows the architecture. Data from six sources enter a collection layer and are stored in a database. The community twin holds the current state of each spatial unit (ward or village): water availability, drainage condition, rainfall, elevation, waste reports and population. AI/ML modules read the twin state, and their outputs (water risk, flood risk and waste detection) are combined by the priority engine, which produces recommended actions for the dashboard. The what-if module modifies twin state variables and re-runs the models to estimate the effect of an intervention.

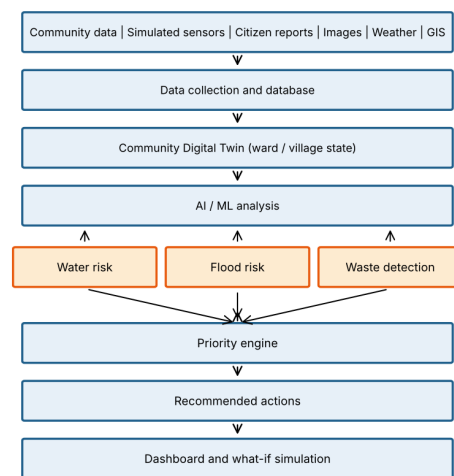


FIGURE 1. CommunityTwin architecture and data flow.

C. Design Principle

Every module follows *Monitor, Understand, Predict, Prioritise, Simulate, Act*. Monitoring and understanding correspond to data fusion in the twin; prediction is performed by the ML modules; prioritisation, simulation and action are decision-support functions exposed through the dashboard. This ordering matters because it separates what is measured, what is inferred and what is recommended. The first stages describe the community as it is, the middle stages estimate what may happen, and only the final stages propose what to do, so that a human administrator can see at which point judgement or assumptions have entered.

D. Data Sources

Simulated sensor data (rainfall, soil moisture, river level, tank level, groundwater depth, leakage) stands in for IoT devices and can be replaced by real streams without changing the models, because the models consume only named feature columns and are indifferent to how those columns are produced. *Citizen reports and images* provide geotagged complaints about water supply, drainage and waste. *Weather and geographical data* provide temperature, rainfall forecasts, elevation and drainage maps. *Historical records* provide past shortages and flood events used as labels.

Table 2 lists the input variables used by the two prediction modules together with the ranges used in simulation. Rainfall and soil moisture are the fast-changing variables that drive the flood module. Elevation and drainage capacity are slowly changing site characteristics. The waste-blockage index, derived from citizen reports, is the variable that couples the waste domain to flood risk. For the water module, population and temperature drive demand, while groundwater depth, tank level and leakage describe supply.

TABLE 2. INPUT VARIABLES USED BY THE PREDICTION MODULES.

Module	Variable	Source	Range (simulated)
Flood	Rainfall 24 h (mm)	Weather / sensor	Gamma(1.2, 14)
Flood	Rainfall 72 h (mm)	Weather / sensor	derived
Flood	Soil moisture	Sensor	0.10-0.95
Flood	Elevation (m)	GIS	2-40
Flood	Drainage capacity index	GIS / reports	0.2-1.0
Flood	River level (m)	Sensor	0.5-6.0
Flood	Waste-blockage index	Citizen reports	0-1
Water	Population	Records	300-5,000
Water	Temperature (°C)	Weather	22-42
Water	Rain 7 d (mm)	Weather	Gamma(1.0, 8)
Water	Groundwater depth (m)	Sensor	2-30
Water	Tank level, leak rate, previous deficit	Sensor / records	0-1, 0-0.3, 0-1

E. Twin State and Update Semantics

The twin is event-driven. Each new sensor reading, forecast or citizen report is validated and written to the database, which updates the state of the corresponding ward or village. The update triggers re-prediction for that unit only, and the refreshed risk scores propagate to the priority engine and dashboard. Because the state of a unit is a small vector, re-prediction is inexpensive (Section IV-L), and the system can respond within the time scale of a rainfall event rather than only in a daily batch. The what-if module operates on a copy of the state so that scenarios never alter the live twin.

F. Twin State Record

To make the notion of a compact state vector concrete, Table 3 lists the fields held for each spatial unit. The record is intentionally small: it holds the latest value of each model input, a handful of derived risk scores, and metadata needed for audit. Static fields (elevation, drainage capacity class, population) change rarely and are loaded from GIS layers and records; dynamic fields are overwritten as new readings arrive, and a time-stamped history is retained in the database for re-training.

TABLE 3. FIELDS OF THE TWIN STATE RECORD FOR ONE WARD OR VILLAGE (DESIGN SPECIFICATION).

Group	Fields	Update frequency
Identity	unit id, type (ward / village), centroid, area	Static
Static context	elevation, drainage capacity index, population, vulnerability index	Rarely (survey / census)
Weather and hydrology	rainfall 24 h / 72 h / 7 d, temperature, soil moisture, river level	Hourly to daily
Water supply	tank level, leak rate, groundwater depth, previous deficit	Daily to weekly
Citizen signals	open reports by category, waste-blockage index, hotspot membership	On each report
Model outputs	water risk, flood probability, waste score, priority score and level	On each update
Audit	last update time, source of each input, model version	On each update

IV. METHODOLOGY

A. Overall Workflow

Fig. 2 shows the workflow applied to each prediction module. Raw inputs from the six data sources are cleaned (range checks, removal of duplicate reports, imputation of missing readings by the last valid value), converted into features, and split into training and test sets. Models are trained with stratified 5-fold cross-validation, evaluated on the held-out set, and then deployed inside the twin, where each new sensor reading or citizen report updates the twin state and triggers re-prediction. The same predictions feed the priority engine and the what-if module.

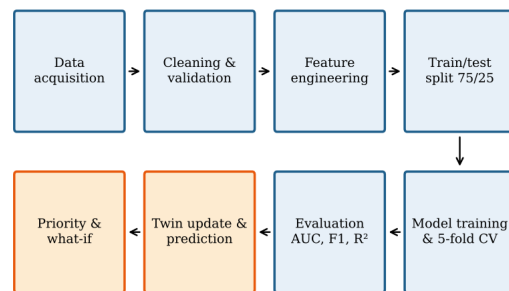


FIGURE 2. Methodology workflow for training, evaluation and deployment of the CommunityTwin models.

B. Data Cleaning and Feature Engineering

Cleaning has three parts. *Range checks* discard readings outside physically plausible limits (for example negative rainfall or a soil-moisture fraction above one). *Duplicate removal* collapses repeated citizen reports of the same issue at the same location. *Imputation by the last valid value* fills short gaps in sensor streams; it is simple, causal (it never uses future information) and easy to explain, although it can bias estimates when gaps are long. Feature engineering is deliberately light. The 72-hour rainfall total is derived from the daily series, and the previous week's deficit is carried forward as a feature for the water module. For the logistic and linear models, inputs are standardised to zero mean and unit variance so that coefficients are comparable in magnitude.

C. Evaluation Protocol and Leakage Control

Care is needed to ensure that reported scores are not inflated by information leakage. The train/test split is made before any model is fitted, and standardisation parameters are estimated on the training portion only. Cross-validation folds are stratified for the flood task so that each fold contains roughly the same share of flood days. The previous-week deficit used as a feature in the water module is taken from the week before the target week and therefore does not contain the target. Hyperparameters are left at library defaults so that no information from the test set can enter through tuning. For real data, the random splits used here would need to be replaced by chronological splits, because consecutive days and weeks are correlated and a random split would leak information from neighbouring records into the test set, giving optimistic estimates of operational performance.

D. Water Shortage Prediction

For each village-week, the model predicts the deficit ratio $D = (\text{demand} - \text{supply}) / \text{demand}$ from population, temperature, 7-day rainfall, groundwater depth, storage-tank level, leakage rate and the previous week's deficit. A village is flagged as a shortage risk when D exceeds the 75th percentile of the observed distribution. We compare linear regression and a random forest regressor [6]. Including the previous week's deficit gives the model a persistence term that reflects how shortages tend to carry over from one week to the next, which is realistic for tank-based supply.

E. Flood Risk Prediction

For each ward-day, a binary classifier predicts flood risk from 24-h and 72-h rainfall, soil moisture, elevation, drainage capacity, river level and a waste-blockage index. In the simulation, the top 20% of a latent hydrological stress score is labelled as flood events. We compare logistic regression, a random forest and gradient boosting [7]. The three models span a range of capacity: logistic regression is linear in its inputs on the log-odds scale, whereas the two tree ensembles can represent thresholds and interactions.

F. Model Descriptions

Linear regression models the deficit ratio as a weighted sum of the standardised inputs plus an intercept, with weights chosen by ordinary least squares. *Logistic regression* models the log-odds of a flood event as a weighted sum of the standardised inputs, and is fitted by regularised maximum likelihood with class weights that are inversely proportional to class frequency, so that the minority (flood) class is not ignored. *Random forests* [6] average many decision trees, each trained on a bootstrap sample with a random subset of features considered at each split; the experiments use 300 trees. *Gradient boosting* [7] builds trees sequentially, each fitted to the residual errors of the ensemble so far. All models use the default hyperparameters of scikit-learn [9]. We did not tune hyperparameters, in order to avoid giving any model an advantage through search effort and to keep the comparison reproducible; tuning is expected to change absolute scores, particularly for the tree ensembles.

G. Model Equations

For completeness, the models take the following standard forms [11]. Let $\mathbf{x}$ denote the standardised feature vector of a record. Linear regression predicts $\hat{y} = b_0 + \mathbf{w}^T \mathbf{x}$, with parameters chosen to minimise the sum of squared residuals. Logistic regression predicts the flood probability

$$p(\text{flood} | \mathbf{x}) = 1 / (1 + \exp[-(b_0 + \mathbf{w}^T \mathbf{x})]) \quad (3)$$

and is fitted by minimising a class-weighted, regularised log-loss. A coefficient w_j has a direct reading: a one-standard-deviation increase in feature j multiplies the odds of a flood by $\exp(w_j)$ when the other features are held fixed. This is the property that lets an administrator see why a ward was flagged. A random forest with T trees predicts by averaging, $\hat{y}(\mathbf{x}) = (1/T) \sum_t h_t(\mathbf{x})$ for regression, or the mean of the trees' class probabilities for classification. Gradient boosting builds an additive model $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + v h_m(\mathbf{x})$, where each new tree h_m is fitted to the negative gradient of the loss at the current ensemble and v is a learning rate. The area under the ROC curve can be written as the probability that a randomly chosen flood day receives a higher predicted probability than a randomly chosen non-flood day [12]:

$$\text{AUC} = \Pr[p(\mathbf{x}_+) > p(\mathbf{x}_-)] \quad (4)$$

which makes explicit why ROC-AUC is insensitive to the alert threshold and to class prevalence, and why it should be reported together with threshold-dependent metrics such as precision and recall.

H. Priority Score

For each ward i , normalised risk scores for water (s_w), flood (s_f) and waste (s_g) are combined with the exposure factor:

$$P_i = (0.35 s_w + 0.40 s_f + 0.25 s_g) (0.6 + 0.25 \text{pop} + 0.15 \text{vul}) \quad (1)$$

where pop is normalised population and vul a vulnerability index. The weights are expert-set defaults and can be adjusted by administrators. Two properties of (1) are worth noting. First, if the three hazard weights sum to one and the scores lie in $[0, 1]$, the hazard term also lies in $[0, 1]$. Second, because

the exposure term ranges from 0.6 (a small, low-vulnerability ward) to 1.0 (the most populous, most vulnerable ward), exposure can raise or lower a ward's score by at most a factor of 1/0.6, about 1.67. Exposure therefore adjusts the ranking without being able to override a large difference in hazard. Flood receives the largest weight because floods are sudden and can endanger life, whereas shortages and waste accumulation typically develop over days or weeks.

A hypothetical example illustrates the effect. Ward A has $(s_w, s_p, s_v) = (0.5, 0.6, 0.3)$, $pop = 0.8$ and $vul = 0.5$; Ward B has $(0.2, 0.9, 0.1)$, $pop = 0.2$ and $vul = 0.1$. Ward B has the larger single risk (0.9), but the combined hazard terms are 0.490 for A and 0.455 for B, and the exposure multipliers are 0.875 and 0.665, giving $P_A = 0.429$ and $P_B = 0.303$. Ward A is therefore escalated first, which is the behaviour the priority engine is designed to produce. The numbers are invented for illustration and are not simulation outputs.

I. Escalation Rules

The priority score is converted into an action level by its percentile across the wards of the community. The rule set used in the prototype is shown in Table 4; it follows the companion study on waste hotspot detection and what-if simulation. Using percentiles rather than absolute thresholds guarantees that a fixed fraction of wards is always flagged for immediate attention, which suits communities with limited field capacity, but it also means that a community in which every ward is safe will still have a nominal 'critical' ward. The level should therefore always be read together with the absolute score.

TABLE 4. PRIORITY LEVEL AND RECOMMENDED ACTION RULES (LEVELS BY SCORE PERCENTILE ACROSS WARDS).

Level	Score percentile	Recommended action
Critical	Top 10%	Immediate field team; notify ward officer
High	75th-90th	Schedule within 48 h (drain cleaning, tanker, extra pick-up)
Medium	40th-75th	Include in weekly work plan
Low	Below 40th	Monitor; no action

With 25 wards, the top 10% corresponds to two or three wards, and the 75th–90th percentile band to roughly four more, so a dashboard user would see a short, manageable action list at each update.

J. What-If Simulation

A scenario is a set of changes to twin state variables, for example drainage capacity +25%, waste blockage removed, or rainfall +50%. The what-if module copies the current (or a representative) state, applies the changes, and re-evaluates the trained flood model on the modified state. The reported quantity is the change in mean predicted flood probability. Because the trained models are statistical, the output indicates the direction and approximate magnitude of an effect rather than a physical simulation: the module reflects associations learned from data, not the hydrodynamics of water moving through a drainage network.

K. Evaluation Metrics

Classification is evaluated with precision (P), recall (R), F1 and ROC-AUC. Regression is evaluated with RMSE, MAE and R^2 :

$$F1 = 2PR / (P + R), RMSE = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}, R^2 = 1 - SS_{res} / SS_{tot} \quad (2)$$

Precision is the fraction of raised alerts that correspond to real floods, and recall is the fraction of real floods that raise an alert. F1 is their harmonic mean and is more informative than accuracy when classes are imbalanced. ROC-AUC measures how well the predicted probabilities rank flood days above non-flood days, independently of any single threshold [12]. For regression, RMSE penalises large errors more heavily than MAE, and R^2 expresses the fraction of variance explained by the model.

L. Interpretability and Cost

Linear models expose coefficients, and tree models expose feature importance, so a non-technical administrator can see why a ward was flagged. The measured inference cost is also low: 0.044 ms per record for the random forest and 0.0004 ms for logistic regression on the test machine (single CPU), which supports the lightweight, software-only design. At these speeds, re-scoring every ward in a community of several hundred units after each sensor update would take a negligible fraction of a second, so the computational budget is dominated by data ingestion and display rather than by the models.

V. EXPERIMENTAL SETUP

All experiments use synthetic data generated with a fixed random seed (42). The flood data set has 6,000 ward-day records (20% positive) and the water data set has 3,000 village-week records. Both sets were split 75/25 (stratified for flood), giving 4,500 training and 1,500 test records for the flood module and 2,250 training and 750 test records for the water module. Models were implemented in scikit-learn [9] with default hyperparameters; logistic regression used standardised inputs and class balancing. Metrics are accuracy, F1, ROC-AUC (flood) and RMSE, MAE, R^2 (water). Table 5 summarises the data sets and model settings.

TABLE 5. DATA SETS AND MODEL SETTINGS.

Item	Flood module	Water module
Records	6,000 ward-days	3,000 village-weeks
Target	Flood event (20% positive)	Deficit ratio D
Split	75/25, stratified	75/25
Models	LR, RF (300 trees), GB	Linear reg., RF (300 trees)
Class weights	Balanced (LR, RF)	N/A
Validation	Stratified 5-fold CV	5-fold CV
Seed	42 (data), 1/2 (split)	42 (data), 2 (split)

Synthetic data were generated from known functions plus noise. Flood events are defined by a latent hydrological stress score, computed from the inputs, whose top 20% are labelled positive. The deficit ratio is likewise computed from the water-module inputs with added noise. The variable ranges in Table 2 define the sampling distributions. This design has one important consequence: because the ground-truth relationships

are known and close to linear, the experiment tests whether the pipeline can recover them, which is a useful software-correctness check, but it cannot tell us how well the models would handle the messy, non-linear relationships of real catchments.

Important note. Because the data are simulated from known functions plus noise, the absolute scores below demonstrate that the pipeline works; they must not be read as performance on real communities.

VI. RESULTS AND DISCUSSION

A. Flood Risk Prediction

TABLE 6. FLOOD RISK CLASSIFICATION ON THE HELD-OUT TEST SET.

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.891	0.766	0.961
Random Forest	0.888	0.717	0.943
Gradient Boosting	0.890	0.682	0.949

Logistic regression achieved the best F1 (0.766) and ROC-AUC (0.961), while the random forest and gradient boosting reached ROC-AUC of 0.943 and 0.949 (Table 6, Figs. 3 and 4). All models reached about 0.89 accuracy. Accuracy is a weak discriminator here because 80% of the test records are non-flood: a classifier that never raised an alert would score 0.80, so 0.89 is better than this trivial baseline but only modestly so, and F1 is the more informative metric. The ordering of the models is not the same on the two metrics. Gradient boosting has a higher ROC-AUC than the random forest (0.949 versus 0.943) but a lower F1 (0.682 versus 0.717), which indicates that its probability ranking is good but that the default 0.5 decision threshold is not well matched to the 20% prevalence. In an operational setting the threshold would be chosen explicitly (Section VII-B) rather than left at its default.

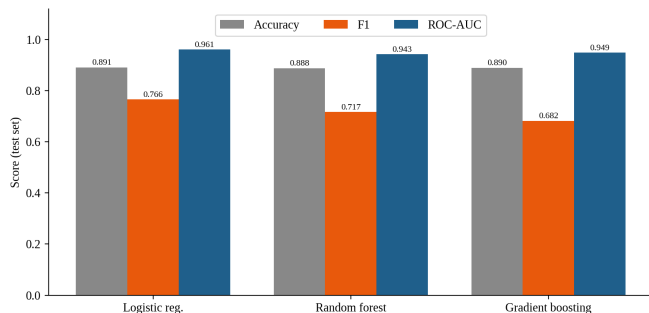


FIGURE 3. Comparison of the three flood models on the held-out test set (values from Table 6).

The linear model is favoured because the simulated stress score is itself a linear function of the inputs; real flood processes are non-linear, and we expect tree-based models to be more competitive on field data. We stress that this is an explanation consistent with the construction of the data, not a finding about hydrology.

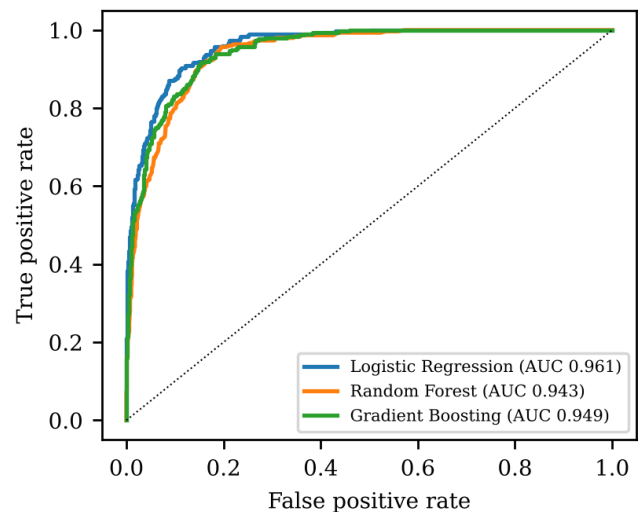


FIGURE 4. ROC curves of the three flood models on the test set.

1) Confusion-matrix analysis.

Fig. 5 shows that logistic regression detected 269 of 300 flood events (recall 0.897) at the cost of 133 false alarms among 1,200 non-flood records, which is acceptable for an early-warning setting where a missed event is costlier than a false alarm. From the counts in Fig. 5 several further quantities follow directly (Table 7): precision is $269/402 = 0.669$, so about two of every three alerts correspond to a real event; the false-positive rate is $133/1,200 = 0.111$; the specificity is 0.889; and the negative predictive value is $1,067/1,098 = 0.972$, so a day on which no alert is raised is very unlikely to be a flood day. The F1 value computed from these counts, $2 \times 0.669 \times 0.897 / (0.669 + 0.897) = 0.766$, matches Table 4.

Actual	No flood	1067	133
	Flood	31	269
		No flood	Flood
		Predicted	

FIGURE 5. Confusion matrix of logistic regression (test set).

TABLE 7. QUANTITIES DERIVED FROM THE CONFUSION MATRIX IN FIG. 5 (LOGISTIC REGRESSION).

Quantity	Definition	Value
True positives / false negatives	Floods detected / missed	269 / 31
False positives / true negatives	False alarms / correct non-alerts	133 / 1,067
Recall (sensitivity)	TP / (TP + FN)	0.897
Precision	TP / (TP + FP)	0.669
Specificity	TN / (TN + FP)	0.889
False-positive rate	FP / (FP + TN)	0.111
Negative predictive value	TN / (TN + FN)	0.972
Balanced accuracy	(recall + specificity) / 2	0.893

2) Cross-validation.

Five-fold cross-validation (Table 8, Fig. 6) gives results consistent with the held-out test set. For flood, logistic regression obtains a mean ROC-AUC of 0.959 ± 0.004 and the random forest 0.946 ± 0.006 ; for water, linear regression obtains $R^2 = 0.964 \pm 0.002$ and the random forest 0.903 ± 0.004 . The standard deviations are small compared with the gaps between models, so the ranking of models is not an artefact of a particular split. Note, however, that this stability reflects the simplicity of the synthetic data-generating process and may not carry over to real, heterogeneous records.

TABLE 8. FIVE-FOLD CROSS-VALIDATION (MEAN $\pm$ STANDARD DEVIATION).

Task	Model	Metric	Mean $\pm$ SD
Flood	Logistic regression	ROC-AUC	0.959 ± 0.004
Flood	Random forest	ROC-AUC	0.946 ± 0.006
Water	Linear regression	R^2	0.964 ± 0.002
Water	Random forest	R^2	0.903 ± 0.004

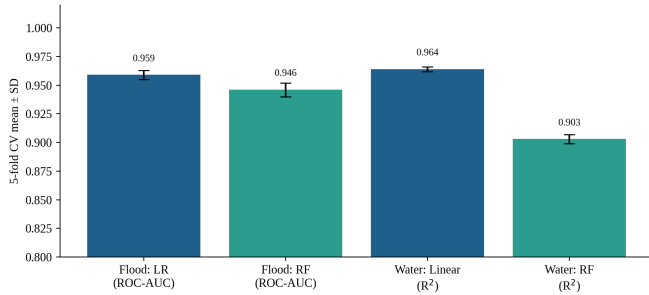


FIGURE 6. Five-fold cross-validation results (mean and standard deviation; values from Table 8).

3) Feature ablation.

The ablation study (Table 9, Fig. 7) removes one feature group at a time from the logistic regression and re-fits the model. Removing rainfall (24 h and 72 h together) caused the largest drop in ROC-AUC (0.961 to 0.865) and in F1 (0.766 to 0.583). Removing river level lowered ROC-AUC to 0.901 and elevation to 0.905; the F1 values after removing these two are almost identical (0.646 and 0.644). Removing the waste-blockage index lowered ROC-AUC only slightly (0.949, a drop of 0.012) and F1 to 0.728, but it is the one variable that links the waste module to flood risk, so it is retained in the twin. Drainage capacity (0.939) and soil moisture (0.936) lie in between.

TABLE 9. ABLATION OF FLOOD FEATURES (LOGISTIC REGRESSION, TEST SET).

Feature set	ROC-AUC	F1	Δ ROC-AUC	Δ F1
All features	0.961	0.766	—	—
– Waste blockage	0.949	0.728	–0.012	–0.038
– Drainage capacity	0.939	0.704	–0.022	–0.062
– Soil moisture	0.936	0.694	–0.025	–0.072
– River level	0.901	0.646	–0.060	–0.120
– Elevation	0.905	0.644	–0.056	–0.122
– Rainfall (24h+72h)	0.865	0.583	–0.096	–0.183

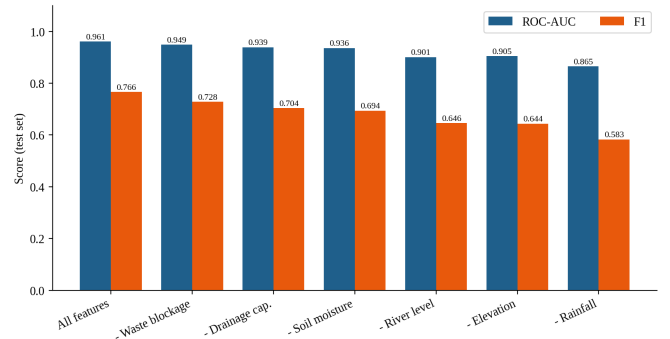


FIGURE 7. Effect of removing each feature group on logistic-regression flood performance (values from Table 9).

The ablation has a practical reading for data collection. Rainfall, river level and elevation carry most of the predictive signal in this simulation, so a community with limited resources should prioritise obtaining reliable rainfall and river-level readings and a good elevation model before investing in finer-grained drainage or waste data. The waste-blockage index contributes the least predictive signal in the simulation, but its value in the twin is structural: it is the channel through which an intervention in the waste domain (clearing blockages) can be represented in the flood domain, and it is needed for the what-if scenarios of Section VI-C.

4) Feature importance.

Fig. 8 shows that river level (0.20), elevation (0.18) and 72-h rainfall (0.17) were the most important features of the random forest, whereas the waste-blockage index ranked last (0.08) in this simulation; 24-h rainfall, soil moisture and drainage capacity lie between roughly 0.10 and 0.16. This ranking differs from the ablation ranking, in which rainfall matters most. The two analyses answer different questions. Forest importance measures how much each feature contributes to the splits of the fitted trees, and when two features are strongly correlated (24-h and 72-h rainfall) the importance is divided between them, which lowers the value of each. The ablation removes both rainfall variables together and measures the loss to a different model (logistic regression). Neither analysis is wrong; together they suggest that rainfall, river level and elevation are the dominant drivers.

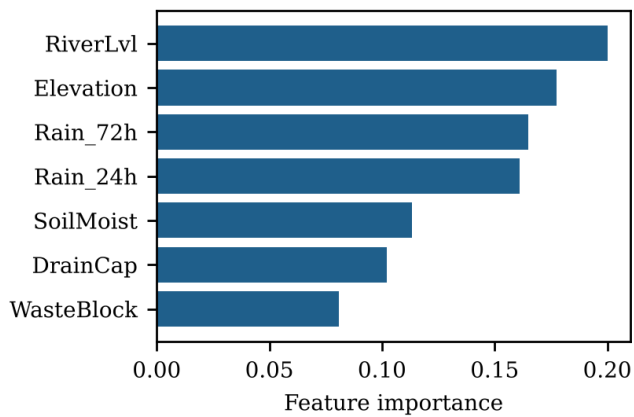


FIGURE 8. Random forest feature importance for flood risk.

B. Water Shortage Prediction

TABLE 10. WATER DEFICIT-RATIO REGRESSION RESULTS.

Model	RMSE	MAE	R ²
Linear Regression	0.041	0.033	0.962
Random Forest	0.066	0.052	0.902

Linear regression achieved RMSE 0.041 and $R^2 = 0.962$; the random forest reached RMSE 0.066 and $R^2 = 0.902$ (Table 10). Since the deficit ratio is on a scale of roughly 0 to 1, an RMSE of 0.041 corresponds to a typical error of about four percentage points of demand. When the predictions are thresholded at the 75th percentile to flag shortage villages, the random forest obtains an F1-score of 0.844. Temperature was the dominant feature in the forest, followed by groundwater depth and population, which agrees with the intuition that heat raises demand and that deeper groundwater and larger populations make a shortage more likely.

Fig. 9 plots predicted against actual deficit ratio for the random forest. Predictions follow the diagonal closely, but the forest appears to compress the extremes, with the highest deficits somewhat underestimated, which is typical of tree ensembles that average over leaves and cannot extrapolate beyond the range of training targets. This behaviour matters operationally, because the villages with the highest deficits are the ones most in need of a tanker. The linear model does not show this compression in the simulation, again because the underlying relationship is linear.

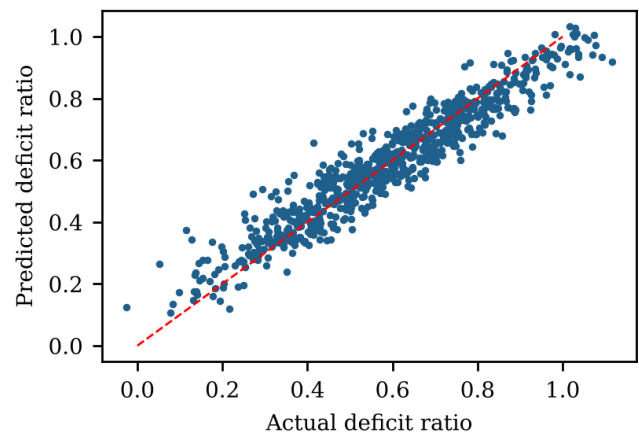


FIGURE 9. Predicted versus actual deficit ratio (random forest).

C. Priority Engine and What-If Simulation

Applying (1) to 25 simulated wards, the priority top-5 list (wards 20, 1, 6, 10 and 22) overlapped with a naive worst-single-risk top-5 list in only 3 of 5 wards, showing that exposure weighting and hazard combination change which locations are escalated. Both lists place ward 20 first. The what-if module re-evaluated the flood model on 1,500 records under two scenarios (Fig. 10): increasing drainage capacity by 25% and removing waste blockage lowered the mean predicted flood probability from 0.259 to 0.155, while a 50% rainfall increase raised it to 0.365.

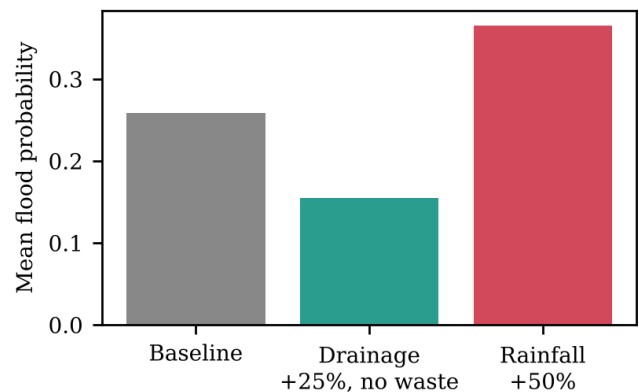


FIGURE 10. What-if simulation of flood probability.

In relative terms (Fig. 11), the combined drainage and waste intervention reduces the mean predicted probability by about 40%, and the rainfall increase raises it by about 41%. The two effects are of similar size and opposite sign: in this simulation, a 25% improvement in drainage together with clearing waste blockage offsets roughly the impact of a 50% increase in rainfall. This is a useful summary for planning discussions but must be read with the caveat in Section IV-J. The directions agree with expectations, which is a basic sanity check of the module, but the magnitudes are properties of the fitted model and the synthetic data.

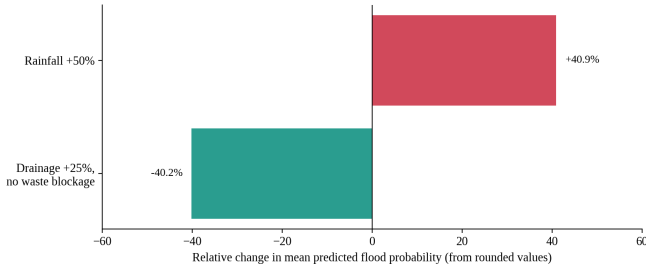


FIGURE 11. Relative change in mean predicted flood probability under the two scenarios, computed from the rounded values in Fig. 10.

Weight sensitivity.

To test how sensitive the ranking is to the expert-set weights, we sampled 500 weight vectors from a Dirichlet distribution centred on the defaults. The mean Jaccard overlap of the top-5 list with the default list was 0.75 (minimum 0.43), and the highest-priority ward stayed the same in 75% of the runs (Fig. 12). The ranking is therefore reasonably stable but not insensitive, which supports exposing the weights to administrators. The three-peaked shape of the histogram follows from the arithmetic of the Jaccard index for two lists of five wards: sharing five, four or three wards gives 1.00, 0.67 or 0.43 respectively, and in none of the 500 runs did the overlap fall below three wards.

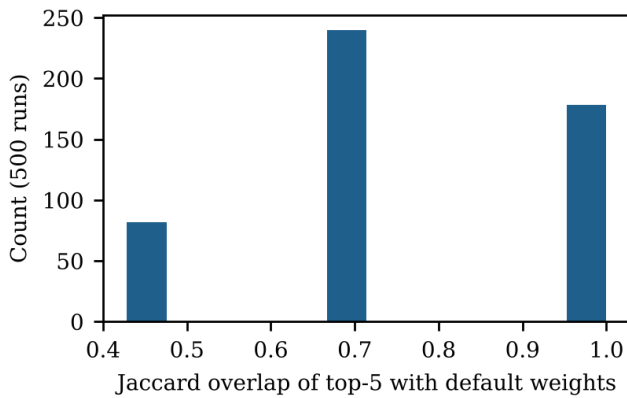


FIGURE 12. Stability of the top-5 priority list under random weight perturbation.

D. Reading the Results Together

Taken together, the results support three modest conclusions. First, the pipeline is internally consistent: the metrics computed from the confusion matrix reproduce the reported F1, cross-validation agrees with the held-out test set, and the scenario effects have the expected sign. Second, in this simulation the predictive signal is concentrated in a small number of variables (rainfall, river level and elevation), which suggests that a minimal sensing package would capture most of the benefit in an analogous real setting. Third, the decision-support layers add information that the prediction layer alone does not provide: the priority engine reorders wards relative to a naive ranking, and the what-if module translates a feature change into a change in risk.

Equally important is what the results do *not* show. They do not show that the models would reach similar scores on field

data, that linear models would outperform tree ensembles in practice, that the priority ranking agrees with expert judgement, or that the what-if magnitudes would be realised physically after an intervention. Each of these claims requires real data and, for the last, a quasi-experimental evaluation of actual interventions.

E. Community Dashboard

Fig. 13 shows an illustrative mock-up of the Community Digital Twin Dashboard: a map of waste hotspots and flood- and water-risk locations, the ranked priority list, a community risk index and a list of recommended actions (for example, clearing drains in ward W20 before forecast rain, sending a water tanker to village V07, and adding a waste pick-up near a hotspot). The dashboard is the point at which model outputs become decisions, so it presents the ranked list and the recommended actions together with the underlying risk scores. The numbers in the mock-up are illustrative and are not outputs of a field deployment.

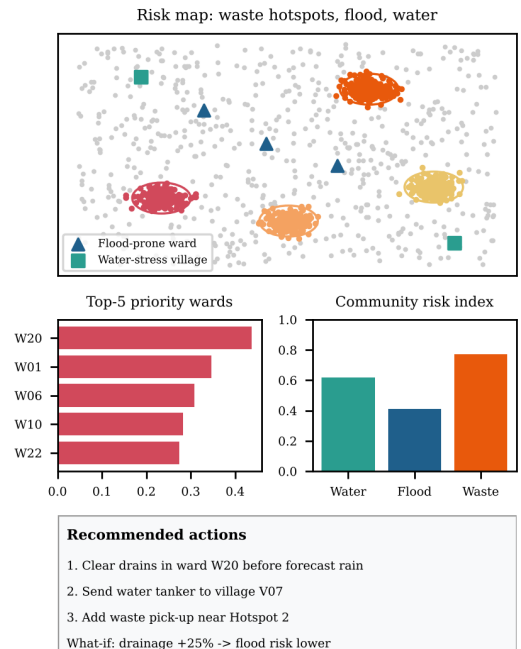


FIGURE 13. Illustrative mock-up of the CommunityTwin dashboard.

VII. DISCUSSION

A. When Simple Models Suffice

A recurring result is that the linear models matched or exceeded the tree ensembles on both tasks. This should not be over-interpreted. It occurs because the synthetic ground truth is close to linear in the inputs, so a model that encodes this assumption has an advantage and needs little data to find the right solution. The result does, however, make a methodological point that is often overlooked in applied machine-learning work: a strong, interpretable baseline should always be established before more complex models are introduced, because it shows how much of the signal is captured by simple structure and prevents the conclusion that a complex model is needed when it is not. For communities with small historical data sets, a well-regularised linear or logistic model may also be more robust than a flexible ensemble.

B. From Probabilities to Alerts

The classifiers output probabilities, but operations require decisions. The threshold at which an alert is raised trades missed events against false alarms. With the confusion matrix in Fig. 5, the chosen threshold yields recall 0.897 and precision 0.669. A community that has few emergency resources might accept fewer alerts and higher precision; a community in which floods cause loss of life would favour higher recall. We recommend that the threshold be a configuration parameter, reviewed with local stakeholders, and that it be tied to the cost of the response action: a low-cost action such as sending an SMS warning justifies a lower threshold than a costly one such as evacuating a settlement.

C. Operational Considerations

Data quality. The twin is only as reliable as its inputs. Sensors drift, citizen reports are subject to reporting bias (areas with higher smartphone penetration generate more reports), and weather forecasts carry uncertainty. The cleaning steps in Section IV-B address only the simplest problems. A deployed system should add sensor health monitoring, uncertainty estimates and checks that flag inputs far from the training distribution.

Human oversight. The twin is a decision-support tool. Recommended actions are suggestions that an administrator can accept, modify or reject, and each suggestion should be accompanied by the evidence behind it (the risk scores, the principal contributing features and the weights used). Interpretable models make this practical.

Privacy and fairness. Citizen reports and images may contain personal information such as faces or house numbers. A deployment should minimise the data it stores, anonymise images, and restrict access. Prioritisation also raises questions of fairness: because the exposure term favours populous and vulnerable wards, but reporting rates differ between wards, a ward whose residents report less may be systematically under-ranked. Monitoring the correlation between report volume and priority rank, and supplementing reports with sensor data, would help detect such effects.

Maintenance. Models trained on historical data can become stale as drainage works are completed, land use changes or the

climate shifts. Periodic re-training, with monitoring of prediction error against observed outcomes, is required. The software-only design makes re-training cheap, since the models are small and training takes seconds.

D. Roadmap Towards Deployment

We envisage a staged path to field use. In the first stage, the twin is run in shadow mode alongside existing practice in one or two communities, with its predictions logged but not acted on, so that they can be compared with outcomes. In the second stage, stakeholder workshops with ward officers, engineers and residents agree on the priority weights, alert thresholds and escalation rules. In the third stage, selected recommendations are acted on, and their results are recorded, which also provides real evidence for calibrating the what-if module. Throughout, real sensor streams replace the simulated ones, and the evaluation protocol of this paper is repeated on real data.

E. Summary of Key Quantitative Results

Table 11 gathers the principal quantitative results in one place for reference. All values are reproduced from the sections above; none is new.

TABLE 11. SUMMARY OF PRINCIPAL RESULTS ON SIMULATED DATA.

Task	Result	Value
Flood (test)	Logistic regression: accuracy / F1 / ROC-AUC	0.891 / 0.766 / 0.961
Flood (test)	Random forest: accuracy / F1 / ROC-AUC	0.888 / 0.717 / 0.943
Flood (test)	Gradient boosting: accuracy / F1 / ROC-AUC	0.890 / 0.682 / 0.949
Flood (test)	LR recall; false alarms among 1,200 non-floods	0.897; 133
Flood (CV)	LR / RF ROC-AUC	0.959 / 0.946
Water (test)	Linear regression: RMSE / MAE / R^2	0.041 / 0.033 / 0.962
Water (test)	Random forest: RMSE / MAE / R^2	0.066 / 0.052 / 0.902
Water (test)	Random forest shortage-flag F1	0.844
Priority	Top-5 overlap with worst-single-risk list	3 of 5
Priority	Top-5 Jaccard under weight perturbation: mean (min)	0.75 (0.43)
What-if	Mean flood probability: baseline / S1 / S2	0.259 / 0.155 / 0.365

F. Coupling with the Waste Module

The flood and water modules in this paper are complemented by a waste module that clusters geotagged citizen reports with DBSCAN [10]. In the companion study on simulated data, DBSCAN recovered all four true hotspots with a recall of 0.994 and a precision of 0.940, and the resulting waste score enters the priority score (1) as s_w . The waste-blockage index in Table 2 is the link in the ^g other direction, from waste to flooding. Representing the coupling explicitly is what allows scenario S1 (clear the blockage and add drainage capacity) to be expressed as a change of state rather than as a separate model.

G. Computational Cost and Scalability

The inference costs reported in Section IV-L allow a simple scaling estimate. At 0.044 ms per record, re-scoring 1,000 wards with the random forest takes about 44 ms of model time on a single CPU; with logistic regression at 0.0004 ms per record the same operation takes about 0.4 ms. These are model-only figures measured on the test machine and exclude database access, feature assembly and network transfer, which are likely to dominate in a deployed system. Even so, the estimate shows that model evaluation is not the bottleneck and that the twin could run on modest hardware, such as a single small server in a municipal office, which is relevant for communities with limited IT resources.

H. Comparison with Alternative Designs

Several alternative designs are conceivable. A *single multi-task neural network* that jointly predicts water, flood and waste risk could share representations between hazards, but it would be harder to interpret and would need considerably more data than a community is likely to have. A *full process-based model* would represent interventions more faithfully but would require surveys of the drainage network and calibration effort that many communities cannot afford. A *rule-based dashboard* that simply displays thresholds on each sensor is cheap and transparent but cannot combine hazards or estimate the effect of interventions. CommunityTwin sits between these: it uses small interpretable models, combines hazards through an explicit scoring rule, and offers approximate scenario estimates. We regard this as an appropriate first step on the path from separate reporting channels to a fully integrated community model.

I. Implementation Considerations

Although this paper evaluates the modelling layer only, several implementation choices follow from the design. The database should store both the current state of each unit and an append-only history of readings and reports, so that models can be re-trained and any recommendation can be reconstructed after the fact. Models should be versioned and each prediction should record the version that produced it. Thresholds, weights and escalation rules should be held as configuration, editable through the dashboard by authorised administrators and logged when changed, because these are the parameters that most directly encode local policy. Finally, input adapters should be isolated from the models: replacing a simulated rainfall stream with a real gauge feed should require changing one adapter, not the model code. These choices are not evaluated here but are consistent with the claim that real streams can replace simulated ones without changing the models.

VIII. ETHICAL, PRIVACY AND SOCIETAL CONSIDERATIONS

Community-scale analytics touch the daily lives of residents, and several issues deserve explicit treatment. *Consent and data minimisation*: citizen reports should be voluntary, should state what will be done with the data, and should collect only what the analysis needs; images should be processed to remove faces and identifying text before storage.

Governance and accountability: responsibility for acting on a recommendation must rest with a named official, and the twin should record who accepted or rejected each recommendation and why. *Over-reliance*: a ranked list can appear more authoritative than it is. The dashboard should display uncertainty and the key assumptions alongside scores, and users should be trained to treat the output as input to judgement. *Equity*: prioritisation rules influence who receives services first. The weights in (1) embody value judgements, for example that floods outrank shortages, and these should be set through an inclusive process rather than by developers alone. *Accessibility*: dashboards and alerts should be available in local languages and in formats usable by officials with limited technical training, and alerts to residents should reach those without smartphones.

J. A Hypothetical Walk-Through

To show how the layers interact, consider a purely hypothetical event; the figures are invented for illustration and are not simulation outputs. A forecast update raises the expected 24-hour rainfall for the catchment that contains ward W20. The twin writes the new forecast to the state of W20 and its neighbours and re-evaluates the flood model. The predicted flood probability for W20 rises above the alert threshold, while those of higher-elevation wards do not. W20 already has a high waste-blockage index from earlier citizen reports and a high vulnerability index, so its priority score from (1) rises sharply and the ward moves into the critical band of Table 4. The dashboard shows W20 at the top of the list with the recommended action to clear drains before the forecast rain. The administrator opens the what-if panel, applies scenario S1, and sees that clearing the blockage and adding drainage capacity would lower the predicted probability for the ward. A field team is dispatched, and when it reports the drains cleared, the blockage index for W20 is set to zero by the new report and the score falls accordingly at the next update. In this sequence the models supply estimates, the scoring rule supplies an ordering, and the administrator supplies the decision, which is the division of labour the architecture is intended to support.

K. Minimum Viable Data Requirements

The ablation and importance results suggest a staged approach to data collection, summarised in Table 12. The grouping is a recommendation derived from the simulated results and should be re-examined with real data: in a real catchment, for example, drainage condition may matter far more than it does in the simulation.

TABLE 12. SUGGESTED STAGING OF DATA COLLECTION, INFORMED BY THE SIMULATED ABLATION (TABLE 9).

Stage	Flood module	Water module
Essential	Rainfall (24 h, 72 h), river level, elevation	Population, temperature, groundwater depth, previous deficit
Valuable	Soil moisture, drainage capacity index	Tank level, leak rate, 7-day rainfall
Structural	Waste-blockage index (needed for cross-hazard what-if analysis)	–

L. Handling Missing and Delayed Data

Real deployments face gaps, delays and conflicting readings that the simulation does not reproduce. Last-value imputation is acceptable for short outages, but the twin should record how stale each input is and degrade gracefully: if a key input such as river level has been missing for longer than a configured period, the dashboard should mark the dependent prediction as low confidence rather than silently reusing the old value. Citizen reports arrive irregularly and with delay, so a ward with no reports is not necessarily a ward with no problems; the absence of data should be displayed as such. Conflicting sources, for example a gauge reading that disagrees with a forecast, should be resolved by explicit precedence rules held in configuration, and large disagreements should be flagged for human review. These measures do not remove uncertainty, but they make it visible to the administrator who must act on the output.

IX. LIMITATIONS AND THREATS TO VALIDITY

The evaluation relies on synthetic data, the priority weights are not calibrated against expert judgement, the what-if module reflects model-learned associations rather than hydrodynamic physics, and the study has not been validated with a real community. These limitations will be addressed through field data collection and stakeholder workshops. We organise the main threats to validity as follows.

Construct validity. The flood label is defined as the top 20% of a latent stress score. In reality, floods are rare events whose frequency varies between wards and years, and their labels (from historical records) are incomplete and noisy. The deficit ratio is likewise defined by a formula rather than measured. Performance on these constructed targets may not reflect performance on real events.

Internal validity. Experiments use a single synthetic data realisation (seed 42), a single train/test split per module, and default hyperparameters. Cross-validation reduces the dependence on a particular split, but we did not repeat the data generation with different seeds, and we did not test the statistical significance of differences between models. Differences of the order of 0.01 in ROC-AUC should not be regarded as established.

External validity. Communities differ in climate, terrain, infrastructure and reporting behaviour. Results obtained for simulated communities with the ranges in Table 2 do not transfer automatically to other settings, and the tree models in particular may behave differently in real data where relationships are non-linear and features are correlated.

Conclusion validity. The what-if magnitudes (a 40% reduction, a 41% increase) are properties of the fitted statistical model. They should not be used as the sole basis for infrastructure investment, for which hydrodynamic modelling and engineering judgement are required. The priority ranking is moderately sensitive to the weights, as shown by the Dirichlet experiment, and the question of whether the combined ranking is closer to expert judgement than a simple ranking could not be assessed without real expert data.

X. FUTURE RESEARCH DIRECTIONS

Several directions follow naturally from the limitations above.

- **Real data and pilot evaluation.** The most important next step is to run the twin in at least one community with real sensor streams and citizen reports, first in shadow mode and then in operation. Appendix E proposes the outcome measures for such a pilot.
- **Uncertainty estimation.** The models currently return point predictions. Calibrated probabilities, prediction intervals for the deficit ratio and conformal or ensemble-based uncertainty would allow the dashboard to show how confident each recommendation is, and would support more careful threshold setting.
- **Temporal models.** Recurrent networks such as LSTM [8] could exploit longer histories and forecast several days ahead. Their value over the simple models used here should be tested explicitly against the linear baseline, in view of the finding that simple models can match tree ensembles when relationships are near linear.
- **Physically informed what-if analysis.** Coupling the statistical models with a simplified hydraulic model of the drainage network would let the what-if module represent interventions that change the structure of the system, such as a new outfall, rather than only shifting model inputs.
- **Participatory calibration.** The priority weights and escalation thresholds should be set through workshops with ward officers, engineers and residents, and re-examined periodically. Methods from multi-criteria decision analysis can structure these discussions.
- **Sensor placement.** The ablation results suggest that a few well-chosen measurements carry most of the signal. Formal analysis of which sensors give the greatest improvement per unit cost would help communities invest sparingly.
- **Fairness auditing.** Because citizen reports are unevenly distributed, regular audits of whether priority ranks track report volume rather than need are advisable.

XI. CONCLUSION

CommunityTwin integrates community data, simulated sensors, citizen reports and weather data into a lightweight digital twin that predicts water shortage and flood risk, ranks wards by priority and supports what-if simulation. On simulated data, the models reached ROC-AUC up to 0.96 for flood risk and R^2 up to 0.96 for water deficit. Ablation and importance analyses identified rainfall, river level and elevation as the dominant drivers of simulated flood risk, the priority engine reordered wards relative to a single-risk ranking, and the what-if module produced directionally sensible estimates of intervention effects. These results demonstrate that the pipeline is feasible, interpretable and inexpensive to run, not that it is accurate in the field.

Future work includes integrating real IoT streams, LSTM-based forecasting [8], uncertainty estimation and a pilot deployment. Further priorities are calibrating the priority

weights through stakeholder workshops, replacing the statistical what-if module with, or complementing it by, a simplified hydraulic model of the drainage network, repeating the experiments over multiple random seeds with significance testing, and evaluating the fairness of the ranking with respect to reporting behaviour.

APPENDIX A: REPRODUCIBILITY SUMMARY

For convenience, the settings required to reproduce the experiments are collected in Table 13. All values are taken from Section V and Table 5; no additional settings were used.

TABLE 13. REPRODUCIBILITY SUMMARY.

Setting	Value
Random seed (data generation)	42
Flood data set	6,000 ward-days; 20% positive; 75/25 stratified split (4,500 / 1,500)
Water data set	3,000 village-weeks; 75/25 split (2,250 / 750)
Flood models	Logistic regression (standardised inputs, balanced class weights); random forest (300 trees, balanced); gradient boosting; scikit-learn defaults
Water models	Linear regression; random forest (300 trees); scikit-learn defaults
Validation	Stratified 5-fold CV (flood); 5-fold CV (water)
Shortage flag	Deficit ratio above the 75th percentile of the observed distribution
Priority weights	Water 0.35, flood 0.40, waste 0.25; exposure 0.6 + 0.25 pop + 0.15 vul
Weight sensitivity	500 Dirichlet-sampled weight vectors centred on the defaults; top-5 Jaccard overlap
What-if scenarios	S1: drainage +25% and waste-blockage index = 0; S2: rainfall (24 h and 72 h) +50%; evaluated on 1,500 records
Inference cost (single CPU)	RF 0.044 ms/record; logistic regression 0.0004 ms/record

APPENDIX B: NOTATION AND ABBREVIATIONS

TABLE 14. NOTATION AND ABBREVIATIONS USED IN THIS PAPER.

Symbol / term	Meaning
D	Deficit ratio, (demand – supply) / demand, per village-week
s_w, s_f, s_g	Normalised water, flood and waste risk scores of a ward
pop, vul	Normalised population and vulnerability index of a ward
P_i	Priority score of ward i , Eq. (1)
ROC-AUC	Area under the receiver operating characteristic curve
F1	Harmonic mean of precision and recall
RMSE, MAE	Root-mean-square error; mean absolute error
LR, RF, GB	Logistic regression; random forest; gradient boosting
DT	Digital twin
DBSCAN	Density-based spatial clustering of applications with noise
BIM, GIS	Building information modelling; geographic information system
IoT	Internet of Things

APPENDIX C: DASHBOARD COMPONENTS

The dashboard shown in Fig. 13 comprises four coordinated components, each tied to a stage of the Monitor-to-Act principle. The *risk map* (Monitor, Understand) plots waste hotspots as clusters and marks flood-prone wards and water-stressed villages. The *ranked priority list* (Prioritise) shows the top wards with their scores. The *community risk index* (Understand) summarises the average normalised water, flood and waste risk across the community, allowing a quick comparison of the three hazards. The *recommended actions panel* (Simulate, Act) lists concrete steps, each linked to the rule in Table 4 that produced it, together with the what-if estimate where one is available. An administrator can open any ward to see its inputs, its risk scores and the weights applied.

APPENDIX D: DEPLOYMENT LOOP

TABLE 15. EVENT-DRIVEN UPDATE LOOP OF THE TWIN (DESIGN SPECIFICATION).

Step	Operation
1	Receive an event: sensor reading, forecast update or citizen report.
2	Validate (range and duplicate checks); impute missing sensor values by last valid value; reject invalid reports.
3	Write the event to the history store and update the state record of the affected unit (Table 3).
4	Re-compute derived features (for example 72 h rainfall, waste-blockage index).
5	Evaluate the deployed flood and water models on the updated state; evaluate the waste score from the current hotspot list.
6	Normalise the three risk scores; compute P_i by Eq. (1) using the configured weights.
7	Re-rank all wards; assign action levels by percentile (Table 4); generate recommended actions for new or escalated wards.
8	Publish the refreshed state, ranking and recommendations to the dashboard; log model versions, weights and thresholds used.
9	On request, copy the state, apply a scenario, re-evaluate the flood model and display the change in predicted risk (the live twin is not modified).

APPENDIX E: PROPOSED PILOT EVALUATION MEASURES

To move from demonstrating feasibility to establishing field value, a pilot should collect the measures in Table 16 and report them against a baseline period or comparison wards. The measures are proposals and have not been applied in this study.

TABLE 16. PROPOSED OUTCOME MEASURES FOR A PILOT DEPLOYMENT.

Aspect	Measure	Purpose
Prediction	Alert recall, precision and ROC-AUC against recorded flood events; deficit RMSE against measured tank shortfalls	Check transfer of simulated performance to field data
Timeliness	Lead time between alert and event; delay from citizen report to action	Show whether the twin shortens response time
Prioritisation	Agreement of the ranking with independent expert ranking; fraction of recommendations accepted	Assess the weights and the usefulness of the list
Interventions	Change in observed flooding or shortage after actions, compared with predicted what-if effects	Calibrate the what-if module
Equity	Correlation between report volume and priority rank across wards	Detect reporting bias
Usability	Administrator trust and workload ratings; time to understand a recommendation	Judge acceptance and interpretability

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